SAFE CARZ

Drive Every Road with Confidence

PROJECT DOCUMENTATION

Vehicle Insurance Marketplace — Pakistan

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Student Introduction

About This Project

Safe Carz is my semester project for the Web Designing module of the AI course at Aptech. The project is a fully functional, responsive, multi-page website that serves as a vehicle insurance marketplace for Pakistani car and bike owners. It was built entirely using frontend technologies — HTML5, CSS3, and JavaScript — with no backend or database involved.

Objective

The main objective of this project was to apply all the web designing concepts learned during the semester — including semantic HTML, CSS animations, JavaScript DOM manipulation, dynamic rendering, form validation, responsive design, and third-party library integration — into one complete, real-world website.

What I Learned

- Building multi-page websites with consistent shared styles using Global.css
- Using JavaScript arrays of objects to store and render data dynamically
- Creating CSS animations using keyframes, transitions, and pseudo-elements
- Implementing responsive design using CSS media queries and flexbox/grid
- Using IntersectionObserver API for scroll-triggered animations
- Integrating third-party libraries — Bootstrap 5, Font Awesome, AOS, Google Fonts
- Implementing client-side form validation with custom success and error alerts
- Building interactive UI components — sliders, filters, sort dropdowns, hamburger menus

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1. Project Overview

What is Safe Carz?

Safe Carz is a fully responsive, multi-page vehicle insurance marketplace built for Pakistani car and bike owners. The platform allows users to browse insurance plans from 6 major companies, filter plans by insurer or vehicle type, sort by price, and contact advisors directly — all from a single website.

Problem Statement

In Pakistan, vehicle owners who want to buy insurance must visit multiple company websites separately, manually compare plans, and often encounter confusing jargon and hidden fees. Safe Carz solves this by centralizing all major insurers on one transparent platform.

Key Features

Technology	Purpose
HTML5	Structure and content of all 6 pages — semantic elements, forms, navigation
CSS3	Styling, animations, keyframes, flexbox, grid, media queries, transitions
JavaScript (ES6)	All interactivity — sliders, filters, dynamic rendering, form validation, hamburger menu
Bootstrap 5	Utility classes — d-flex, align-items-center, justify-content-between, gap in navbar

Font Awesome 6	Icon library — phone, map marker, check circle, social media icons throughout all pages

- Dynamic plan filtering by insurer and vehicle type with real-time results count
- Price sorting — Low to High and High to Low
- Hero image slider with auto-play and manual navigation
- Scrolling ticker banner showing key services
- Animated plan cards with shimmer hover effects
- Company profiles with statistics and plan listings
- Contact form with JavaScript validation and success/error alerts
- Fully responsive design with hamburger navigation on mobile
- Scroll-triggered fade animations using IntersectionObserver API
- AOS (Animate On Scroll) library on the About page
- Google Maps integration on Contact page
- 6 Insurance Companies — Zurich, EFU General, Adamjee, Jubilee, TPL, Salama

2. Technologies Used

Technology	Purpose
AOS Library	Animate On Scroll — scroll animations on About page using dataaos attributes
Google Fonts	Poppins font family — weights 300, 400, 600, 700 for clean modern typography
Global.css	Shared stylesheet linked to all pages — navbar, buttons, hamburger, responsive rules

3. Page 1 — Home Page

File: index.html | Style: HomePage.css | Script: Homepage.js

Purpose

The homepage is the entry point of Safe Carz. Its purpose is to immediately communicate what the platform offers, build user trust through social proof and partner logos, and direct users to explore plans or contact advisors.

Sections & Features

Hero Slider

A full-screen image slider with 3 slides. Each slide contains a background image, a dark rgba overlay for text readability, a tag label, heading, paragraph, and two call-to-action buttons. The slider is built in JavaScript using querySelectorAll to select all slides. The goTo() function removes the active class from the current slide and adds it to the target slide. CSS transition: opacity 1s handles the smooth fade effect. setInterval() auto-plays the slider every 5000ms. Each active slide text animates using CSS keyframes — tag uses slideInTag, heading uses fadeInLeft, buttons use fadeInUp.

Ticker / Marquee Banner

A continuous scrolling banner on a dark red background showing key services like Comprehensive Insurance, 48-Hour Claims, and Nationwide Coverage. Built using pure CSS keyframes animation called scroll that moves the ticker-inner div from translateX(0) to translateX(-50%) infinitely at 24 seconds. Items are duplicated to ensure seamless looping.

Why Choose Us — 6 Feature Cards

6 cards addressing user concerns — Instant Quote, Fast Claims, Expert Advisors, Transparent Pricing, 24/7 Support, Nationwide. Cards use a shimmer hover effect via CSS ::before pseudoelement sliding across the card on hover. Cards lift on hover using transform: translateY(-8px) scale(1.01). Fade-in on scroll using IntersectionObserver API.

Partners Section

Displays logos of all 6 partner companies — Zurich, EFU, Adamjee, Salama, Jubilee, TPL. Logos are grayscale by default. On hover they become full color with scale(1.1) transform.

Plans Preview Section

Shows 4 sample plan cards giving users a price preview before exploring further. View All Plans button links to Plans.html.

Feature Section

Two-column CSS flexbox layout — image on left, mission text on right. Image has hover scale(1.03) effect. Quote block has animated left border on hover.

CTA Section

Dark red background with floating Email Us and phone number buttons animated using CSS keyframes floatY. Three info boxes showing address, email, and working hours with hover lift effects.

JavaScript Functions

Function / Feature	Description
goTo(n)	Navigates slider to slide index n. Updates active class and dot indicators.
move(dir)	Calls goTo(current + dir). dir is +1 for next, -1 for previous.
setInterval	Calls move(1) every 5000ms for auto-play.
IntersectionObserver	Watches .fade elements, adds .show class when entering viewport.
hamburger click	Toggles .open class on hamburger and navlinks for mobile dropdown.

4. Page 2 — Companies Page

File: companies.html | Style: Companies.css | Script: Companies.js

Purpose

Allows users who have a preferred insurer to go directly to that company profile and view all their plans. Also helps undecided users compare companies based on experience, rating, and plan offerings.

Sections & Features

Sticky Filter Bar

7 pill-shaped filter buttons. CSS position: sticky top: 64px. On click, activeCompany variable updates and applyFilters() runs. applyFilters() hides all company panels then shows only the matching one using display:block.

insuranceData — Array of Objects

All 6 companies stored in JavaScript array. Each object has: id, name, badge, description, image, stats {experience, claimTime, rating}, and plans array. Each plan has: name, make, type, tag, price, features array.

renderAllCompanies() Function

Dynamically generates all HTML. Loops through insuranceData using forEach. For each company generates plan cards using nested map() with template literals, then generates the full company panel and injects into the DOM using innerHTML +=.

JavaScript Functions

Function / Feature	Description
applyFilters()	Shows all panels if activeCompany is all. Otherwise hides all and shows only matching panel.
renderAllCompanies()	Loops through insuranceData, generates HTML for each company panel, injects into DOM.
hamburger click	Toggles .open class. Runs after renderAllCompanies() completes.

5. Page 3 — Plans Page

File: Plans.html | Style: Plans.css | Script: Plans.js

Purpose

The core page of Safe Carz. Gives users complete control to find their ideal plan by filtering 36 plans simultaneously by company and vehicle type, with price sorting.

Sections & Features

Dual Filter System

Two filter rows — Row 1 filters by insurer (7 options), Row 2 filters by vehicle type (3 options). Both work simultaneously. Clicking updates activeInsurer or activeType and calls renderCards(). Active buttons get gold underline via CSS border-bottom.

Results Bar & Sort

Live count of matching plans. Sort dropdown: Default, Price Low to High, Price High to Low. onchange calls sortPlans() which calls renderCards().

PLANS Data Array

36 plans stored in JavaScript array. Each object: insurer, type (four-wheeler/two-wheeler), price (number), badge, company, sub, name, make, tag, features array.

renderCards() Function

(1) Removes existing cards. (2) Filters PLANS by activeInsurer and activeType. (3) Sorts by price if selected. (4) Calls buildCard() for each. (5) Inserts into grid. (6) Updates count. (7) Shows no-results if empty.

Card Entrance Animation

CSS keyframes cardEnter: 0% opacity:0 translateY(32px) scale(0.97), 60% bounce, 100% settled. Each card has staggered delay = index * 0.06s.

JavaScript Functions

Function / Feature	Description
renderCards()	Filters PLANS array, sorts if needed, builds and injects cards, updates count.
buildCard(p, index)	Creates a plan card DOM element from plan object with animation delay.

sortPlans()	Called by sort dropdown. Calls renderCards() which handles sorting internally.
tab-btn click	Updates activeInsurer or activeType then calls renderCards().
hamburger click	Toggles .open class on hamburger and navlinks.

6. Page 4 Vehicles Page

File: Vehicles.html | Style: Vehicles.css | Script: Vehicles.js

Purpose

Organizes insurance plans by vehicle type. Targets users who think in terms of their vehicle — showing all plans for four-wheelers and two-wheelers separately with full feature transparency.

Sections & Features

Filter Bar

3 buttons — All Vehicles, Four Wheelers (car icon), Two Wheelers (motorcycle icon). Icons from Font Awesome. Updates activeCat and calls renderAll(). renderAll() adds/removes hidden-block class on section divs.

VEHICLES Data Array

42 vehicle plan objects. Each has: cat (four-wheeler/two-wheeler), company, sub, badge, name (vehicle model), make (engine/year), tag, price, features array. Each feature: {text: string, inc: boolean}. inc:true = included, inc:false = not included.

buildCard() Function

Creates card dynamically. If feature.inc is true — Font Awesome check-circle icon in dark red. If false — times-circle icon in gray with (Not Included) text. Full transparency about what is and is not covered.

JavaScript Functions

Function / Feature	Description
buildCard(v, index)	Creates vehicle plan card. Shows check/cross icons based on feature.inc boolean.
renderAll()	Filters VEHICLES by cat, clears grids, builds and injects cards for both sections.

tab-btn click	Updates activeCat and calls renderAll().
IntersectionObserver	Watches .fade elements and cards. Adds .show class on intersection.
hamburger click	Toggles .open class on hamburger and navlinks.

7. Page 5 About Page

File: About.html | Style: About.css | Script: About.js

Purpose

Builds credibility and trust. Before purchasing insurance users need to know who Safe Carz is, its founding story, growth milestones, and track record.

Sections & Features

Hero Section

Full screen hero with background image. Tag, heading, paragraph animate using CSS keyframes heroSlideIn with staggered delays of 0.2s, 0.4s, 0.6s — starting from opacity:0 translateY(20px).

Our Story Section

Two-column CSS grid layout. Uses AOS data-aos=fade-right on text and data-aos=zoom-in on image. Tells founding story — Safe Carz started 2020 by Ahmed Raza in Karachi PECHS.

Timeline Section

5 milestones from 2020 to 2024. CSS ::before creates vertical center line in dark red. Each card uses data-aos=fade-up with incremental delays 100ms to 500ms. Cards have hover lift: translateY(-4px).

Statistics Section

Dark red background, 4-column CSS grid: 15,000+ Clients, 8 Insurer Partners, 48hr Claims, 20+ Cities. Numbers in Georgia serif gold font.

AOS Library

AOS.init({ duration: 900, once: true, easing: ease-out-quart, offset: 100 }). duration:900ms, once:true means animation plays only once per element.

JavaScript Functions

Function / Feature	Description
AOS.init()	Initializes Animate On Scroll with duration 900ms, once:true, easeout-quart easing.
smooth scroll	Intercepts anchor clicks, uses scrollIntoView({behavior:smooth}).
hamburger click	Toggles .open class on hamburger and navlinks.

8. Page 6 Contact Page

File: Contact.html | Style: Contact.css | Script: Contact.js

Purpose

The conversion point of the entire platform. After browsing plans users need a direct way to take action. Provides contact form, office information, and Google Maps integration.

Sections & Features

Hero Section

Same staggered heroSlideIn animations as About page with 0.2s, 0.4s, 0.6s delays.

Contact Form

Fields: Full Name, Email (2-col grid row), Phone, Subject dropdown, Message textarea. All inputs have focus states — border-color transitions to dark red on focus. Subject dropdown options: Get a Quote, File a Claim, Plan Inquiry, General Question.

Form Validation

addEventListener on submit with e.preventDefault() stops page reload. Checks name, email, message using .trim(). If empty: errorAlert shows (display:block), hides after 4000ms via setTimeout. If valid: successAlert shows, form resets with this.reset(), hides after 5000ms.

Contact Sidebar

CSS position:sticky top:120px. Shows address, email, working hours. Info rows have hover translateX(6px). Icon circles scale(1.15) rotate(12deg) on hover. Open in Google Maps button uses Maps search API URL.

JavaScript Functions

Function / Feature	Description
form submit handler	e.preventDefault(). Validates name/email/message. Shows success or error alert.
setTimeout()	Hides error after 4000ms and success after 5000ms automatically.
this.reset()	Clears all form fields after successful submission.
IntersectionObserver	Watches .fade elements, adds .show class on viewport intersection.
hamburger click	Toggles .open class on hamburger and navlinks.

9. Global CSS & Shared Components

File: Global.css — linked to all 6 pages

Purpose

Global.css contains all styles shared across every page following the DRY principle — Don't Repeat Yourself. Instead of writing navbar and hamburger code 6 times, it is written once and linked to all pages.

Breakpoint	Effect
max-width: 1100px	Plans grid: 3 columns to 2 columns
max-width: 991px	Navbar links hidden, hamburger shown. Nav height 80px to 64px. Logo shrinks.

max-width: 900px	Company intro 2-col to 1-col. Footer 4-col to 2-col.
max-width: 767px	Hero height reduces. Why cards 3-col to 2-col. CTA buttons stack vertically.

Feature	Description
CSS Reset	margin:0, padding:0, box-sizing:border-box on all elements.
Body styles	Poppins font family, color, background white.
Typography	h2 Georgia serif 34px, p 15px line-height 1.8 font-weight 300.
.label	Uppercase small-caps label with dark red color and ::before bar element.
.btn-red	Dark red background button with hover effect.
.btn-white	Transparent white border button with gold hover.
nav	Fixed position, height 80px, white background, box-shadow, flexbox.
.hamburger	Hidden by default. Three span bars that animate to X on .open class.
@media max-width:991px	Shows hamburger, hides nav-right, navlinks become vertical dropdown.
@media max-width:480px	Smaller nav height 58px, smaller logo.

10. Responsive Design

Approach

All 6 pages use CSS media queries. Hamburger navigation handled globally in Global.css. Page-specific layout changes in each page CSS file.

Breakpoints

Breakpoint	Effect
max-width: 700px	Plans grid 1-col. Vehicles padding reduces. Hero padding reduces.
max-width: 600px	Companies padding reduces to 20px. Story section padding reduces.
max-width: 480px	Nav height 58px. Logo further shrinks. Cards 1-col. Filter labels hidden.

Hamburger Menu

Hamburger button has 3 span elements. navlinks has max-height:0 overflow:hidden by default. JavaScript toggles .open class — max-height becomes 400px with 0.4s transition creating smooth dropdown. Spans animate: span1 rotates 45deg, span2 opacity:0, span3 rotates -45deg — forming an X. Clicking any link removes .open class to close menu.

11. JavaScript Functions Reference

Function / Feature	Description
goTo(n) — Homepage	Navigates hero slider to slide n. Updates active class and dot indicators.
move(dir) — Homepage	Calls goTo(current + dir). +1 for next, -1 for previous.
setInterval — Homepage	Calls move(1) every 5000ms for auto-play slider.
applyFilters() — Companies	Shows/hides company panels based on activeCompany variable.
renderAllCompanies() — Companies	Generates all company HTML from insuranceData array.
renderCards() — Plans	Filters, sorts, and renders plan cards from PLANS array.
buildCard(p,i) — Plans	Creates single plan card DOM element with animation delay.
sortPlans() — Plans	Called by sort dropdown. Calls renderCards() internally.
renderAll() — Vehicles	Renders four-wheeler and two-wheeler cards into separate grids.
buildCard(v,i) — Vehicles	Creates vehicle card. Shows check/cross icons based on feature.inc.
AOS.init() — About	Initializes AOS with duration:900, once:true, ease-out-quart.
smooth scroll — About	Intercepts anchor clicks, uses scrollIntoView({behavior:smooth}).
form submit — Contact	e.preventDefault(), validates, shows alerts, resets on success.
setTimeout — Contact	Hides success after 5000ms and error after 4000ms.
IntersectionObserver — All	Watches .fade elements, adds .show class on viewport entry.
hamburger click — All	Toggles .open class on hamburger spans and navlinks dropdown.

12. References

The following resources were used during the development of this project:

No.	Resource	Type	URL
1	Bootstrap 5	Official Documentation	https://getbootstrap.com/docs/5.3
2	Font Awesome 6	Icon Library	https://fontawesome.com
3	AOS Library	Animate On Scroll	https://michalsnik.github.io/aos
4	Google Fonts	Poppins Font Family	https://fonts.google.com/specimen/Poppins
5	MDN Web Docs	HTML, CSS, JavaScript Reference	https://developer.mozilla.org
6	W3Schools	Web Development Tutorials	https://www.w3schools.com
7	CSS Tricks	CSS Flexbox & Grid Guide	https://css-tricks.com
8	Google Maps API	Maps Search Integration	https://developers.google.com/maps
9	cdnjs	Content Delivery Network for Libraries	https://cdnjs.cloudflare.com
10	Aptech Learning	Course Material & Lectures	https://www.aptech-education.com

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